Enhancing Article Classification through Data Exploration and Feature Engineering

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***Abstract*—In this report, we present a supervised classification pipeline for news articles. The dataset presents significant challenges due to the imbalance in the class distribution and the presence of raw HTML artifacts and unstructured text. Our approach focuses on a specialized preprocessing strategy that extracts semantic information from HTML tags, such as image descriptions, rather than simply discarding them. Furthermore, we leverage the high "purity" of specific news sources through a targeted text boosting strategy. We benchmark a Support Vector Machine classifier model (SVM), exploiting Stochastic Gradient Descent (SGD). The model is trained on hybrid character and word-level TF-IDF features against a naive baseline and other linear models. The proposed method demonstrates that careful cleaning, metadata exploitation, and robust feature engineering are critical for performance in web-scraped datasets, achieving a macro F1-score of 0.741 on the public test set.**

I. Problem Overview

The objective of this project is to develop a robust classification model capable of categorizing news articles into seven distinct classes: International News (0), Business (1), Technology (2), Entertainment (3), Sports (4), General News (5), and Health (6). The provided dataset consists of 79,997 labeled samples for training and development, and 20,000 additional samples for evaluation. Each entry includes metadata such as the source, title, timestamp, and PageRank score, along with the raw article content.

A. Data Distribution and Characteristics

An exploratory data analysis was conducted to understand the underlying structure of the dataset. As illustrated in Figure 1, the class distribution is not uniform. While the dataset is relatively balanced, certain categories, such as International News (Class 0), are overrepresented compared to others. This imbalance necessitates careful consideration during model training to prevent bias towards the majority classes.

Another critical observation concerns the lexical similarity between categories. We analyzed the number of shared terms among the top 100 most frequent words for each class. As depicted in the heatmap in Figure 2, International News (Class 0) exhibits a notable overlap in the vocabularies of both General News (Class 5) and Entertainment (Class 3). This substantial intersection suggests a semantic proximity that may introduce ambiguity, complicating the classification task for models relying solely on simple bag-of-words representations.

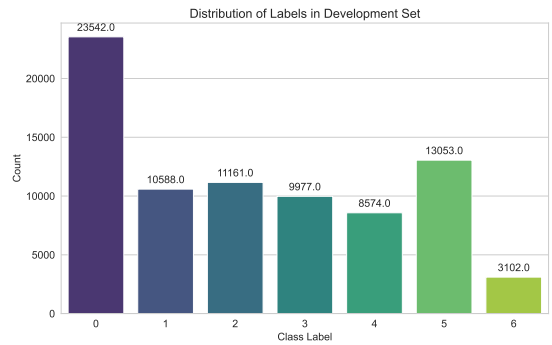

Fig. 1. Distribution of class labels in the development set, highlighting the imbalance across categories.

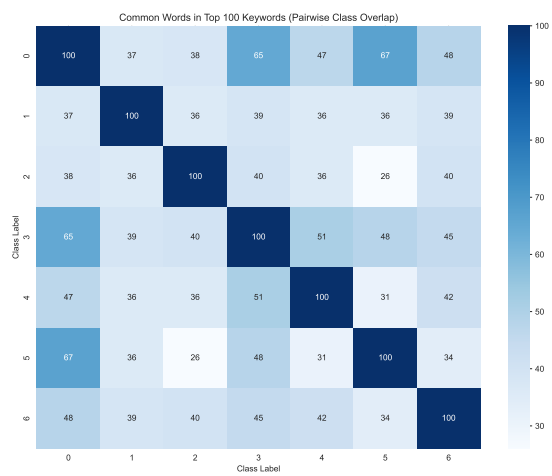

Fig. 2. Pairwise intersection of the top 100 most frequent words across classes. The heatmap highlights significant lexical overlap between specific category pairs, particularly Class 0 and Class 5.

Additionally, we analyzed the distribution of article lengths (word count) across different categories to identify potential discriminative features. The length of the articles varies significantly by topic, as detailed in Table I. For instance, Technology articles (Class 2) are generally the longest, with a mean length of approximately 44.8 words and a high standard deviation (88.6), indicating significant variability and the presence of unusually longer articles (max 1905 words). Conversely, General News articles (Class 5) are the shortest

on average (24.9 words) with the lowest maximum length (77 words). This distinct variation suggests that article length is a valuable feature for differentiating between certain classes.

TABLE I
STATISTICS OF ARTICLE WORD COUNTS ACROSS DIFFERENT CATEGORIES.

Label	Mean Length	Std Dev	Max Length
0 (International News)	37.27	23.46	403
1 (Business)	27.75	16.67	184
2 (Technology)	44.76	88.63	1905
3 (Entertainment)	34.74	21.10	290
4 (Sports)	35.72	18.15	243
5 (General News)	24.91	12.00	77
6 (Health)	32.46	16.28	190

Finally, we investigated the relationship between news sources and article categories. Our analysis revealed that certain publishers demonstrate a high degree of topic specificity. As illustrated in Figure 3, specific sources are almost exclusively associated with a single category (e.g., 'ESPN' with Sports, 'PCWorld' with Technology). This high "purity" implies that the 'source' attribute contains significant predictive signal. Consequently, incorporating source information, potentially with increased weight or through specific feature engineering, is likely to substantially enhance the classifier's performance for these subsets of data. Moreover, the total amount of the samples that contains the selected "pure_sources" is around 10,000 ($\backsim 12\%$ of the development set).

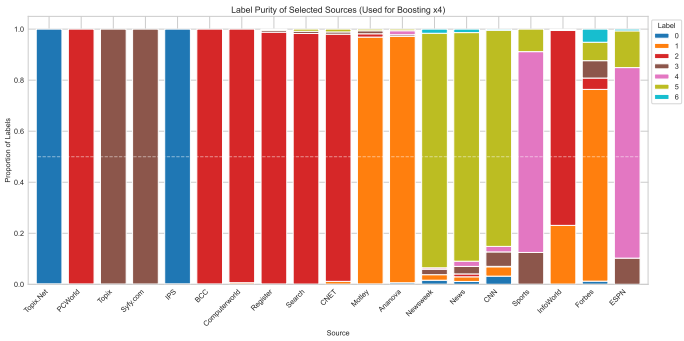

Fig. 3. Label purity of selected news sources. High purity indicates that a source predominantly publishes articles of a single category, making it a strong predictive feature.

II. PROPOSED APPROACH

A. Data Preprocessing and Feature Engineering

Given the noisy nature of the dataset and the importance of specific attributes identified in the data analysis, we implemented a robust feature engineering and feature selection pipeline.

Text Boosting Strategy: To leverage the high predictive power of certain sources (as verified by the purity analysis), we adopted a "text boosting" strategy [1]. This approach consists in selectively replicating metadata fields within the input representation, thereby increasing their relative term frequency weight with respect to the article body. Given the combinatorial nature of the repetition factors and the associated computational cost, the final configuration was selected through an empirical tuning process on the validation set, restricting the search to a feasible subset of candidate combinations. The resulting configuration is:

- *Matched "Pure" Sources:* If the source matches one of the identified high-purity publishers, we construct the input as:
  $source \times 4 + title \times 4 + article \times 2$.
- *Standard Sources:* For all other sources, the construction is:
  $source \times 2 + title \times 3 + article \times 2$.

Temporal and Structural Features: We engineered auxiliary features to capture temporal patterns and document structure:

- *Temporal Features:* We extracted the day of the week (0-6) and the moment of the day. To better generalize and avoid overfitting, we used relaxed binning for the time of day:
  - Morning to Lunch (1): 06:00 to 14:59
  - Afternoon to Evening (2): 15:00 to 20:59
  - Night (3): 21:00 to 05:59
- *Length Features:* We calculated the character length of both the article and the title. To focus on the order of magnitude rather than raw counts, these values were scaled using a logarithmic transformation ($\log(1 + x)$).

Vectorization and Preprocessing Pipeline: The final feature extraction pipeline is implemented using a *ColumnTransformer* that processes distinct data types in parallel:

1) *Source and Temporal Encoding*: The *source* attribute and *temporal* features are encoded using a OneHotEncoder with a minimum frequency threshold of 50 to handle rare categories.
2) *Text Vectorization:* We utilize TF-IDF vectorization [2] with a *SnowballStemmer* to map words to their roots, reducing the size of the vocabulary and capturing semantic similarity. This is followed by a *SelectKBest* step to retain the most informative features.
3) *Character N-Grams:* A separate TF-IDF vectorizer captures character-level n-grams, which helps in detecting sub-word patterns and robustness to misspellings. This is also filtered through *SelectKBest*.
4) *Numerical Features:* The PageRank score and the log-transformed length features are standardized using *StandardScaler*.

B. Model Selection

We evaluated several classification algorithms, including Random Forest and Support Vector Machines (SVM) [3]. We explore different types of SVM with LinearSVC [4], and SGDClassifier. The *SVM* with *SGDClassifier* (Stochastic Gradient Descent) [5] was selected as the optimal model for this task. It offers superior efficiency on large-scale sparse data [6](such as our high-dimensional TF-IDF vectors) and provides flexibility through various loss functions (e.g., 'modified_huber' for probability estimation) and regularization terms ('alpha'), making it highly adaptable and robust against overfitting.

C. Hyperparameter Tuning

To optimize the model's performance, we performed a grid search with 5-fold cross-validation on the development set. The search space included parameters for both the preprocessing steps (feature selection limits) and the classifier itself. The detailed grid search space is defined in Table II.

TABLE II
GRID SEARCH HYPERPARAMETERS SPACE

Parameter	Values
Preprocessing Component	
Word $max_features$	$\{60k, 80k, 100k, 120k\}$
Word SelectKBest k	$\{40k, 50k, 60k, 70k\}$
Char $max_features$	$\{40k, 50k, 60k\}$
Char SelectKBest k	$\{20k, 25k, 30k\}$
OHE $min_frequency$	$\{30, 50, 70\}$
SVM (SGDClassifier)	
$loss$	{log_loss, modified_huber, hinge}
$alpha$	$\{10^{-6}, 10^{-5}, 5 \cdot 10^{-5}, 7 \cdot 10^{-5}, 10^{-4}\}$
$penalty$	{l2, elasticnet}
SVM (LinearSVC)	
C	$\{0.1, 0.5, 1.0, 2.0\}$
$loss$	{hinge, squared_hinge}

III. RESULTS

The experimental evaluation was conducted using the Macro F1-score as the primary metric, consistent with the competition's evaluation criteria. This metric is particularly appropriate for multiclass classification on imbalanced datasets, as it treats all classes equally, preventing the majority class from dominating the performance assessment.

Table III presents a comparative summary of the models evaluated. We established a baseline performance using a naive approach, which yielded a Macro F1-score of 0.443. Subsequently, we explored more complex architectures. The Random Forest classifier demonstrated a substantial improvement, achieving a validation score of 0.689, thereby highlighting the importance of non-linear interactions and ensemble learning.

However, given the high dimensionality of the feature space resulting from the TF-IDF vectorization (100,000+ features), linear models proved to be more effective. The Linear Support Vector Classifier (LinearSVC) surpassed the Random Forest with a score of 0.703.

Ultimately, the proposed SVM with SGDClassifier, enhanced with our specific feature engineering pipeline (Text Boosting and Hybrid Character-Word N-grams), achieved the best performance. It recorded the highest validation score of **0.726**. Crucially, this model demonstrated excellent generalization capabilities, as evidenced by the Public Score of **0.741** on the unseen test set. This confirms that the combination of "pure" source boosting and robust regularization effectively captured discriminative patterns without overfitting.

A. Error Analysis

To gain deeper insights into the model's performance, we analyzed the confusion matrix (Figure 4). The matrix reveals that the classifier performs exceptionally well on distinct categories such as Technology (Class 2) and Sports (Class 4).

TABLE III
PERFORMANCE COMPARISON (MACRO F1-SCORE)

Model	Validation Score	Public Score
Naive Baseline	0.443	0.443
Random Forest	0.689	-
SVM (LinearSVC)	0.703	-
SVM (SGDClassifier)	**0.726**	**0.741**

However, a closer inspection identifies a specific area of confusion: the model tends to misclassify International News (Class 0) as General News (Class 5) and vice versa. Moreover, the model struggled to correctly classify articles belonging to Entertainment (Class 3). This behavior is perfectly consistent with our previous data analysis findings on lexical similarity. As shown previously in Figure 2, these categories share a significant portion of their top vocabulary. This semantic overlap creates a natural ambiguity that even a robust model like SVM struggles to fully resolve, confirming that the shared vocabulary is a primary source of classification error for these specific classes.

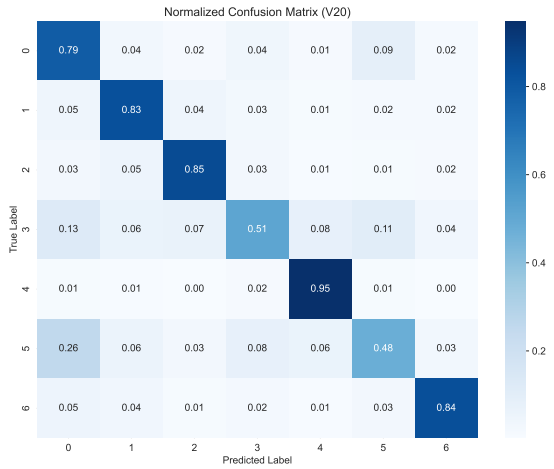

Fig. 4. Normalized Confusion Matrix of the final SGDClassifier. Note the off-diagonal values between Class 0 and Class 5, reflecting the lexical overlap discussed earlier.

IV. DISCUSSION

The results obtained in this study demonstrate the effectiveness of combining domain-specific feature engineering with robust linear classification methods for news categorization.

A key finding of our work is the significant impact of the "source boosting" strategy. By capitalizing on the high purity of specific publishers (e.g., heavily weighting 'ESPN' for Sports), we effectively transformed a difficult text classification problem into a much simpler metadata-based routing task for a subset of the data. This explains the model's high precision in categories like Sports and Technology. Furthermore,

the integration of character-level n-grams proved instrumental in capturing latent morphological structures and handling the noise, allowing the model to capture patterns even when word-level tokens were malformed.

However, the persistent confusion between International News (Class 0), General News (Class 5) and Entertainment (Class 3) highlights a fundamental limitation of bag-of-words approaches. As our lexical analysis showed, these categories share a substantial portion of their vocabulary. Despite optimizing the decision boundary effectively, the model struggles to disambiguate context, as it relies solely on term frequency rather than semantic understanding.

In conclusion, while semantic overlap remains a challenge, the proposed solution achieves a strong balance between efficiency and accuracy (0.741 Public Macro F1), demonstrating that careful preprocessing and metadata exploitation can yield competitive results even without computationally expensive deep learning architectures.

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